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Estimation of Smearing Damage Risk in Wind Turbine Main Bearings

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ABSTRACT

In the global quest for sustainable energy, wind energy is a key renewable resource. The pursuit of lower costs for energy drives continuous innovation in wind turbine technology. Advances in drivetrain design have enabled higher capacity factor and improved turbine performance. Larger wind turbines have led to increasing demands on rolling bearings. Main bearings in particular must support greater loads as rotor diameters grow. Design standards focus on rolling contact fatigue and plastic deformation as primary failure mechanisms. However, damage such as adhesive wear also occurs in practice and can lead to wind turbine failure. Smearing damage in main bearings is an underrepresented failure mode in current design practices, despite its occurrence in the field. This study introduces a simulation-based methodology to assess and reduce smearing risk under realistic conditions. A multibody simulation model is developed to model the main bearing. Specifically, the model consists of a double-row tapered roller bearing with a pin-type cage. Due to the cage's size, the model is extended to account for its flexibility. A screening study with realistic dimensions and load conditions identifies critical scenarios. Roller kinematics and contact forces are analyzed to reveal the damage mechanisms. In one specific load case, pronounced deceleration of the rolling elements is observed, caused by the roller being clamped between the cage and the ring riban effect not represented in existing quasi-static tools. A corrective measure increased preload is implemented and evaluated for effectiveness. The results show a significant reduction in the damage indicator without exceeding fatigue stress limits, demonstrating that the presented approach is capable of extending bearing life and improving system reliability.

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Introduction

Wind turbines are designed for a lifespan of over 20 years (2019), and rolling bearings are expected to meet this based on standard calculations like ICE 61400-4, (1) which references bearing design standards DIN ISO 281 and DIN ISO 76. However, in practice, gearbox bearings sometimes fail much earlier. Main bearings also show premature failures, yet they have been underrepresented in research and reliability studies, often grouped under general drivetrain categories. (2–7) This lack of attention may be due to historically lower reported failure rates. As turbines grow in size, main bearings failures may be increasing, but there is currently insufficient public data to confirm this trend. A significant knowledge gap remains in understanding main bearing failures, including data, modeling, and design standards—especially compared to the extensive research available on gearboxes and generators. (8) DIN ISO 281 deals with dynamic load ratings and rated life calculations for conventional rolling bearings where life is predominantly dependent on rolling contact fatigue. (9) DIN ISO 76, on the other hand, is used for design against plastic deformation. (10) Main bearings of wind turbines operate under dynamic bending moments. Several simulations are necessary to cover the range of load

situations. For large-scale bearings, structural deformation needs to be taken into account in the design process. Therefore, FEA is often utilized for their design. Besides several research models (11) some bearing manufacturers have developed their own FEA tools for bearing calculation. (12) However, the standard design process does not cover damage mechanisms such as wear, adhesive wear, or electrical damage. As adhesive wear is also reported as a failure mechanism in wind turbine main bearings, there is a need to find analysis and prediction methodologies for this type of damage. (6) The causes of adhesive wear, including smearing and rehardening of the surfaces, have been well researched for bearings of size about $d \approx 160$ mm. Harmful slippage is one of the causes. However, slip states cannot be reliably predicted using established methods. Some of them occur during time-dependent events and therefore require transient simulation methods, such as dynamic simulations. The elasticity and resulting deformations of the cages must not be ignored. In order to include cage stiffness in a MBS model of rolling bearings, Hahn presented a method for deep groove ball bearings connecting rigid cage segments with calculated stiffness matrices from an FE model. (13) This approach is successfully applied in Fiedler et al. (14) for modeling a more complex brass cage and a

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Nomenclature

A	contact area (mm ²)	n	rotational velocity (rpm)
b	contact width (mm)	N	number of rows (-)
C_{0t}	roughness-specific factor (-)	p	pressure (MPa)
c_{Rz}	lubricant film roughness factor (-)	P	specific frictional power (mW/mm ²)
d	inner diameter of bearing (mm)	$\bar{R}_z$	mean roughness depth (μm)
d_p	pitch diameter (mm)	t	time (s)
d_R	roller diameter (mm)	t_{2b}	time needed, by the roller, to roll the contact width (s)
D	outer diameter of bearing (mm)	t_{2ba}	time point of beginning roller acceleration (s)
$\vec{F}$	contact force (kN, N)	t_{2ea}	time point of ending roller acceleration (s)
$\bar{F}$	applied load (N)	$\vec{u}$	deformation of a segment (mm)
F_x	force in x direction (kN)	$v_{c,R}$	circumferential velocity (mm/s)
F_y	force in y direction (kN)	v_{rel}	relative velocity (mm/s)
F_z	force in z direction (kN)	v_s	sliding velocity (mm/s)
F_F	total frictional force in contact (N)	v_r	rolling velocity (mm/s)
F_N	contact force (kN)	w	bearing width (-)
h_{min}	minimal film thickness (μm)	z	number of rollers (-)
i	index of slices (-)	Δb	width of an element in the rolling direction (mm)
k	number of slices (-)	Θ_R	moment of inertia of a rolling element (kg/m ²)
$\vec{K}$	segment stiffness matrix (N/mm)	μ	coefficient of friction (-)
l_R	roller length (mm)	λ	asperity load ratio (-)
M_x	torque about x axis (kNm)	τ	threshold stress (MPa)
M_y	torque about y axis (kNm)		
M_z	torque about z axis (kNm)		

three-part window cage for spherical roller bearings. For model verification the deformation of the segmented cage in MBS and a full FE cage model were compared. Furthermore, Büscher implemented a segmented cage model in a commercial MBS software package (15) on the example of a pitch bearing and extended the model for detailed contact calculation between roller and raceways. (16) Yuqing et al. used a lumped-mass flexible cage like the one developed by Hahn in their investigations on the skidding dynamic performance of rolling bearings under acceleration. (17) Detailed bearing models for specific applications have been developed by Sadeghi et al. to identify conditions in deep groove ball bearings that lead to unstable cage behavior. They focused on grease-lubricated rolling bearings and investigated the influence of lubrication condition and the lubricant supply in the cage pockets by varying coefficient of friction and loss factors. (18–21) However, no studies have yet been published on the main bearings of wind turbines, let alone suitable models that are publicly available. Only the rolling bearing manufacturers have access to such models. (22–24) Some of them show measures to avoid damage. (25,26) In addition to bearing manufacturers' instruction for preventing surface damage risks, characteristics for predicting damage due to adhesive wear in particular have been published. Table 1 gives an overview of the most commonly used parameters and the limit values obtained from bearing tests. Similar parameters, based mainly on single contact analysis, have also been published in the English-speaking world. However, for the sake of clarity, they are not mentioned here.

In Table 1, p is the contact pressure, v_{rel} the relative velocity, v_s is the sliding velocity, v_r the rolling velocity, and μ the coefficient of friction. Depending on the literature, this may refer to only the friction in direct contact with the solids (26) or to the total coefficient of friction. (28,29) The asperity

load ratio λ quantifies the load share transmitted via roughness peaks. It contributes to shear stress generation within the contacting surfaces. The solid body thrust stresses in Koch and Vierneusel (26) are modified by a threshold stress τ . The term Δb is the width of an element in the rolling direction. The time needed, by the roller, to roll the contact width is t_{2b} . F_F in Wadewitz (30) and Eglinger (31) is meant as the total frictional force in contact. The terms t_{ba} and t_{ea} are the time of the beginning and end of the acceleration process. These characteristic damage indicators have all been determined for small bearings or at single contact test rigs. Based on the measurement technology developed by Hiltcher (32), Hambrecht conducted damage tests on grease-lubricated roller bearings. (33) He employed the moment of inertia of the rolling elements Θ_R , circumferential velocity $v_{c,R}$, contact area A , and a roughness-specific factor $C_{0t} = \frac{h_{min}}{c_{Rz} \cdot \bar{R}_z}$. All listed damage indicators assume a relationship between roller slippage and risk of damage. As in other aspects of main bearing design and damage assessment, the number of published studies is limited. (6)

The occurrence of smearing damage is primarily governed by the relative sliding velocity in combination with the local contact pressure. In particular, detailed knowledge of the kinematics within the inlet zone of the contact is essential. Consequently, dynamic simulations are required to accurately capture the transient contact behavior and assess the risk of smearing under realistic operating conditions.

Smearing damage assessment

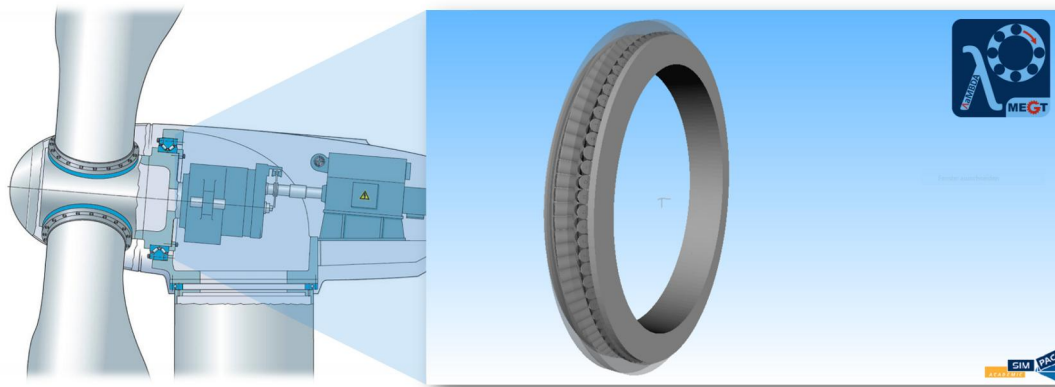
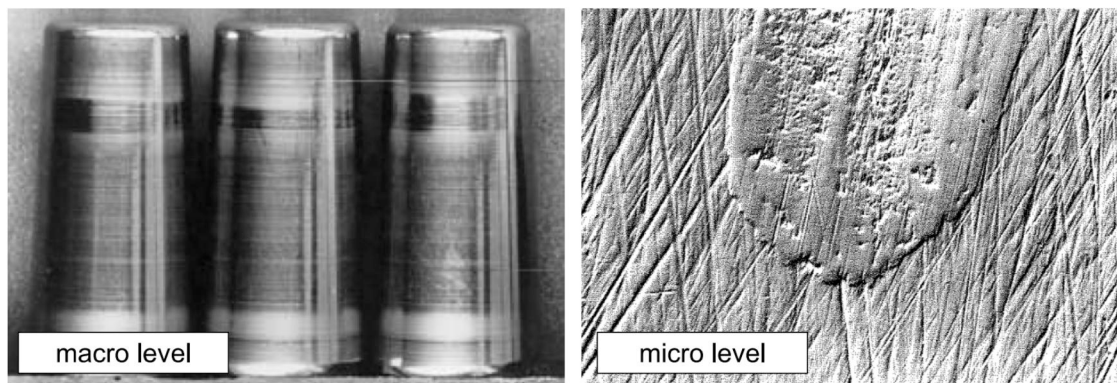
As outlined in the introduction, there are currently no publicly available tools for considering smearing damage in the design process of main bearings for wind turbines. This study presents a methodology for systematically

Table 1. Characteristic damage indicators for smearing and surface initiated damages.

Parameter	Equation	Limit value	Reference
Specific power	$p \cdot v_{rel}$	$300 \frac{W}{mm^2}$	(27)
Specific frictional power ^a	$p \cdot v_{rel} \cdot \mu$	$344 \frac{mW}{mm^2}$	(28)
Smearing criterion	$\max(0.5 \cdot p_{max} \cdot v_{rel} \cdot \mu)$	$30 \frac{W}{mm^2}$	(29)
Local frictional energy	$\int_t^{t+tb} F_f \cdot v_{rel} dt$	$10...15 mJ$	(30)
Sum of frictional energy	$\int_{t_{ba}}^{t_{ba}} \int_t^{t+tb} F_f \cdot v_{rel} dt dt$	$150...250 mJ \cdot ms$	(30)
Specific frictional energy	$\int_t^{t+tb} \frac{F_f \cdot v_{rel}}{2 \cdot b} dt$	$0.35...0.45 \frac{J}{mm}$	(31)
Damage risk value ^b	$\frac{1}{K} \sum_{i=1}^K (p_i \cdot \lambda_{ij} \cdot \mu_{ij} \cdot \tau_{u,i} \cdot v_{s,i} \cdot \frac{\Delta b}{v_{r,i}})$	–	(26)
Specific energy load	$\int_t^{t+tb} \frac{\frac{\partial p}{\partial g}(\dot{\omega}(t) \cdot v_{c,R}(t) \cdot \mu(t) \cdot F_N(t) \cdot v_{rel}(t))}{A(t) \cdot C_{qt}(t)} dt$	$100 \frac{mJ}{mm^2}$	–

^aValue K calculated on the basis of a roller mass of $m_R \approx 8kg$ and a number of rollers per row of $z = 91$ and a specific limit value of $K_{Gr} = K \cdot m_R/n = 30N/mm^2 \cdot m/s \cdot g$.

^bNo specific value obtained.

**Figure 1.** MBS model of a double-row tapered roller bearing used as a main bearing in a wind turbine created on basis of a LaMBDA model.**Figure 2.** An example of smearing damage on macro (25) and micro level. (34)

incorporating this type of damage into the bearing design process. To achieve this, the dynamic simulation tool LaMBDA, developed at the Chair of Machine Elements and Gear Technology (MEGT), is extended to enable the simulation of wind turbine main bearings. Depending on the drive train design, these bearings may be configured as double-row tapered roller bearings with either pin-type or window cages. Since pin-type cages were not previously available in LaMBDA, they were implemented into the simulation framework for this study.

Using an academic example based on realistic bearing dimensions and load conditions, a procedure is established for assessing the risk of smearing. Four representative load

cases, reflecting actual wind-induced forces acting on a main bearing, are analyzed. For a predefined bearing preload, the resulting smearing risk is evaluated. Finally, a mitigation measure aimed at reducing the damage risk is developed and its effectiveness is demonstrated.

The case examined concerns an offshore turbine in which the rotor shaft is supported by a double-row tapered roller bearing and connects to the gearbox. A schematic representation of such a system is visualized in Fig. 1 as an example. As shown in the two examples in Fig. 2, smearing damage may not appear to be severe. However, it is often the initial cause of further, more serious damage to the bearings, which can subsequently lead to failure of the entire system.

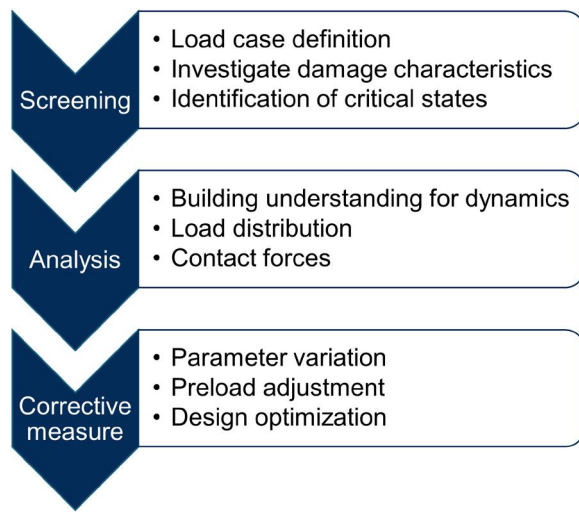


Figure 3. Flow chart of the procedure for damage cause assessment with regard to the identified damage mechanism smearing.

Table 2. Example load case table.

	LC01	LC02	LC03	LC04
Load and unit	Extreme	Standard	Idle	Minimum
F_x [kN]	39	−48	0	0
F_y [kN]	−1944	1852	−1985	−956
F_z [kN]	−533	−247	0	0
M_x [kNm]	5940	−8413	6281	1090
M_y [kNm]	−9358	2207	0	0
M_z [kNm]	0	0	0	0

Methods

Smearing damage often occurs where the rolling element slippage is significant, that is, when a lot of energy is introduced due to high relative speeds in contact with a high load. The loads and kinematics that lead to such conditions are often time-dependent. Transient simulations are therefore necessary to determine the causes. Accordingly, the present study employed dynamic simulation.

Approach

LaMBDA is a combination of software components that allows MBS models of rolling bearings to be set up and simulated in the commercial MBS software Simpack, which was developed in Wingertszahn et al. (35) The TRB models have been validated against friction torque measurements and other calculation tools in several studies. (36–38) The tool has also been utilized for the evaluation of smearing risks and damage analysis. (39–41) These investigations included small-size TRBs with comparatively stiff cages ($d \approx 160$ mm). Therefore the cages were modeled rigid and the influence of cage deformation was neglected. In this study, the model was extended so that the deformation of the cage is also modeled (see the section *Cage flexibility*). In the approach used here, the dynamics simulation was used in several steps. The procedure is shown in Fig. 3.

In the first step, a screening study was carried out. Several load cases, derived from real applications, are simulated and analyzed with regard to the risk of smearing

Table 3. Geometry of the main bearing.

Parameter	Variable	Value	Unit
Outer diameter	D	3300	mm
Pitch diameter	d_p	2900	mm
Inner diameter	d	2500	mm
Number of rollers	z	91	–
Number of rows	N	2	–
Bearing width	w	240	mm
Roller diameter	d_R	95	mm
Roller length	l_R	130	mm
Nominal pressure angle	α	45	°
Roller cone angle	κ	2.654	°
Cage type		Pin type	

damage. The loads are listed in Table 2. In this study, a rotational speed was prescribed instead of an external torque, as a full drive train model was not available. Accurately defining the torque would have required detailed information on the rotor and gearbox dynamics, which were not accessible. Therefore, the rotational speed was used as a boundary condition to approximate the system behavior while focusing on the bearing-level analysis. The rotational speed of $n = 12$ rpm was applied throughout the simulation.

- LC01—Extreme: A high yaw moment acts on the main bearing due to strong, asymmetric wind loading. This represents an extreme operating scenario with high structural loading on the drivetrain.
- LC02—Standard: Wind shear leads to higher wind speeds at the top of the rotor compared to the bottom, inducing a pitch moment on the main bearing. This case reflects common operating conditions under partial rotor asymmetry.
- LC03—Idle: The rotor rotates at a constant speed ($n = 10$ rpm) without significant external loads or moments. This simulates an unloaded coasting condition, such as during low wind or grid disconnection.
- LC04—Minimum: A temporary unloading of the rotor weight occurs due to vertical wind gusts or flow effects, resulting in minimal bearing load.

Critical conditions were identified from the simulation results. The underlying load cases are analyzed in detail in the *Analysis* section to gain an understanding of the bearing kinematics that lead to high damage indicators. The effects of the individual contacts on the rolling elements were analyzed and the resulting acceleration and deceleration processes of the rolling elements were evaluated. A corrective measure was then developed based on the condition of the rolling element kinematics (see *Corrective measure* section).

MBS model

The bodies were modeled according to the geometry in Table 3 (see Fig. 1). For contact calculation, high-resolution contact models from LaMBDA are utilized for the roller-raceway contact. (36) Roller-rib contact is modeled using an integral model as described in Wingertszahn et al. (35)

- Roller-raceway: slice model (alternative slicing technique) with detailed EHL friction.

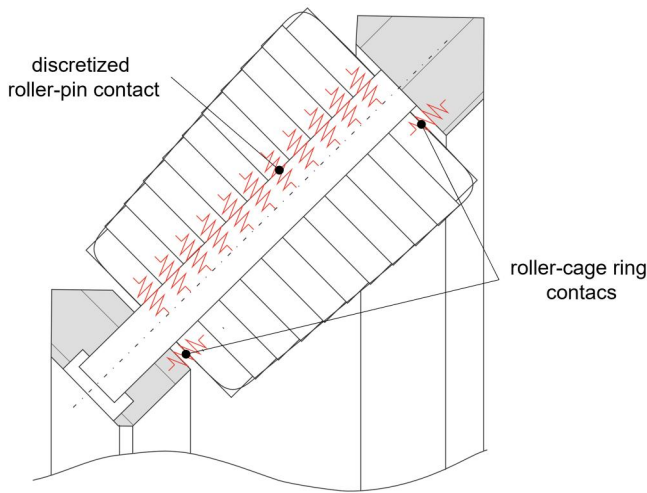


Figure 4. Schematic representation of the pin-type cage integrated into LaMBDA, showing the modeled contacts between the rolling element and the pin, as well as between the rolling element and the cage rings.

- Roller-rib: integral model (Hertzian) with cubic friction model $\mu_1 = 0.02$ at $v_1 = 0.001$ m/s, $\mu_2 = 0.01$ above $v_2 = 0.100$ m/s.
- Roller-pin: slice model (Hertzian) with coulomb friction model $\mu = 0.08$ at $v = 1$ m/s.
- Roller-cage ring: integral Hertzian with coulomb friction model $\mu = 0.02$ at $\omega = 0.1$ rad/s.

Two contacts are relevant for modeling the cage shown in Fig. 4: the axial contact between the rolling element end face and the cage rings, and the contact between the rolling element and the pin. The axial contact is implemented using a simplified contact model that considers frictional effects with the use of a speed-dependent coefficient of friction. The contact between pin axis and the bore in the rolling element is modeled similarly to the raceway contact. A one-dimensional discretization is realized in disks.

As this contact does not need to be examined in detail for the analysis of the damage, only a small number of support points were used for the discretization. The contact force is modeled as a line contact between two cylinders according to Hertz and friction is calculated based on Coulomb's law. The double-row tapered roller bearing model is assembled from the modified single-row bearing by combining two submodels and the intermediate ring.

Cage flexibility

The cage represents a thin-walled structure, and as such, its deformation under self-weight must be taken into account in the simulation. In order to consider the cage's macro-elasticity, the approach developed by Hahn is applied. (13) Thereby, the macro-elasticity is described by discretizing the cage into n segments, with n being the number of rollers guided by the cage. Each segment supports one roller. The segments are in serial connection using spring-damper (SD) elements attached to nodes at their cage pocket center. The

nodes are located in the center of mass of each cage segment. The force elements include two 6×6 matrices, one containing the stiffness values and the second the damping values. The stiffness values are determined from an FE model of a cage segment. To determine the stiffness matrices, an FE model of an intermediate segment is derived from the CAD data of the cage and meshed. The segment model is held in place via the nodes at the interfaces. The nodes at the second intersection are used to apply a load. This is defined as a unit load and is applied in six steps as a force or moment in all spatial directions. The segment stiffness $\vec{K}$ is calculated from the quotient of the deformation of the segment $\vec{u}$ and the applied load $\vec{F}$ using Eq. [1]. This results in a 6×6 matrix with the stiffness in the direction of the coordinate axes of the reference coordinate system on the main diagonal and the coupling terms on the secondary diagonals:

$$\vec{K} = \vec{F} \cdot \vec{u}^{-1} \quad [1]$$

On the other hand, the contact forces are calculated numerically according to Eq. [2]:

$$\vec{F} = \vec{K} \cdot \vec{u} \quad [2]$$

The structural damping is coupled with the stiffness; therefore, as is often simplified, (13,15,17) it is assumed that the damping entries can be estimated by scaling the stiffness matrix with a damping value. The thus-obtained SD elements are implemented in the dynamic simulation to include the deflections of the cages resulting from the weight and contact forces inside of each cage pocket.

Results and discussion

The results are presented according to the procedure in Fig. 3. First, the simulated load cases are analyzed with regard to a damage indicator in the section *Screening*. Based on this, critical load situations are determined and then analyzed in depth (see the section *Analysis*). Subsequent to acquiring an in-depth understanding of the bearing kinematics and the cause of the damage, a corrective measure is developed and its success is evaluated in a new simulation cycle (see the section *Corrective measure*). Finally, the effect of the corrective measure on the remaining load cases is evaluated in the section *Influence of the preload increase on standard load cases*.

Screening

A previous study (42) has shown that the specific frictional power value corresponds very well with the more complex indicators to be calculated from Table 1. It is equally suitable for determining the damage location and also for the risk itself. For this reason, the specific frictional power according to Hansberg (28) was used in the screening study carried out here. Since the focus is on the risk of smearing and this type of damage normally occurs on rolling elements

and raceways, as shown in Fig. 5, the damage parameter for this contact was evaluated. The results of the screening study are shown in Fig. 5.

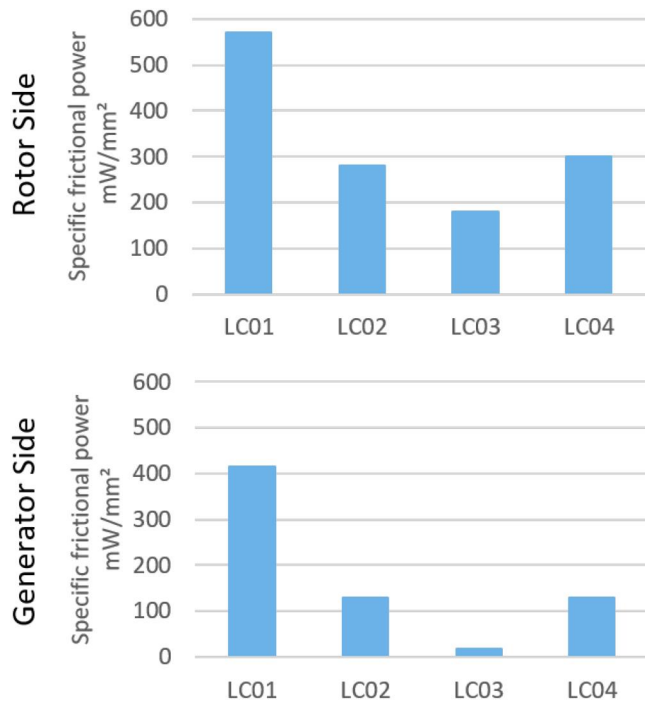


Figure 5. Screening of load cases with regard to the risk of smear damage.

Two findings can be derived from the screening results. Load case LC01 shows by far the highest characteristic values. Second, the values in each of the four simulated load cases are significantly higher on the rotor side than on the generator side. To understand these results, load case LC01 is analyzed in detail as follows.

Analysis

Rolling element slip is critical for smearing damages, and thus, a thorough analysis of the kinematics is carried out prior to the development of a corrective mechanism. As illustrated in Fig. 6, the roller speed and the contact forces in the bearing for load case LC01 are shown over the entire circumference of the bearing for one rotation. Additionally, the contact forces at specific rolling element positions, namely, 2 and 4, are visualized in detail.

At position 1, the rolling element exits the load zone, where it initially decelerates significantly due to the reduced load. Consequently, the frictional forces act on the rolling element without the acceleration provided by the inner ring. This deceleration is characteristic of pin-cage guided bearings of this size. Following this deceleration, a brief acceleration occurs at position 2 due to contact with the inner rib. The rolling element gets into contact with the raceway and the inner rib simultaneously due to gravity. The friction at

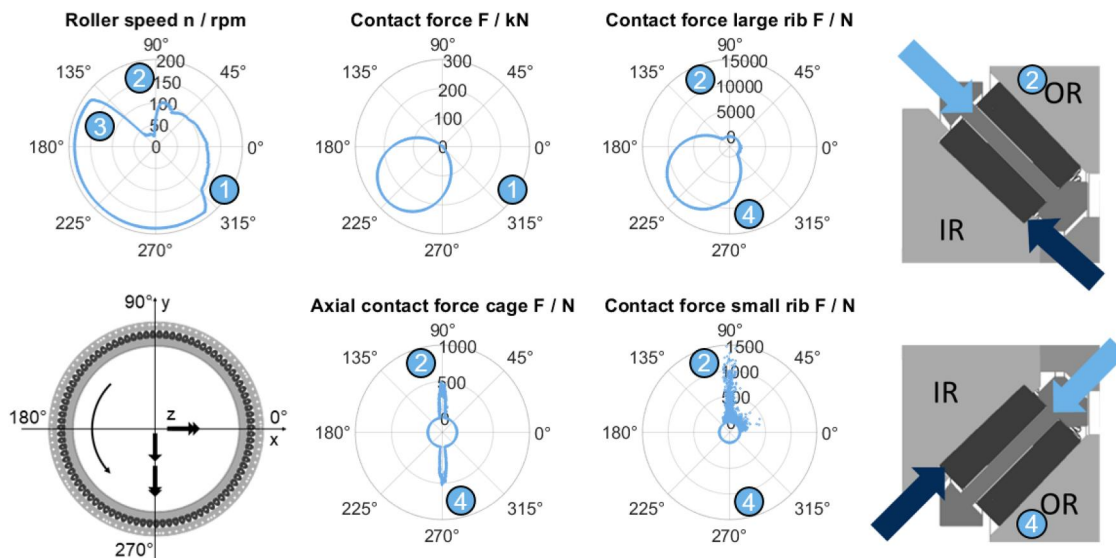


Figure 6. Roller speed, load situation, load zone, contact force between roller and outer rib, contact force between roller and inner rib, contact force between roller and cage ring (axial contact), and contact situation at roller positions 2 and 4 (right) at rotor side of the main bearing.

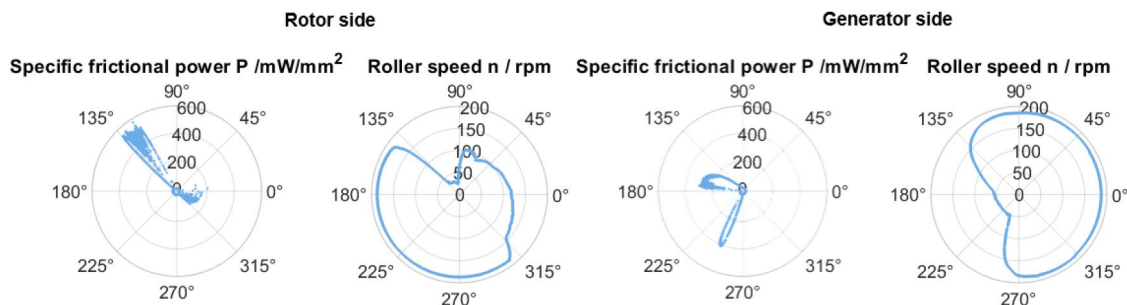


Figure 7. Damage indicator and roller speed at rotor side (left) and generator side (right).

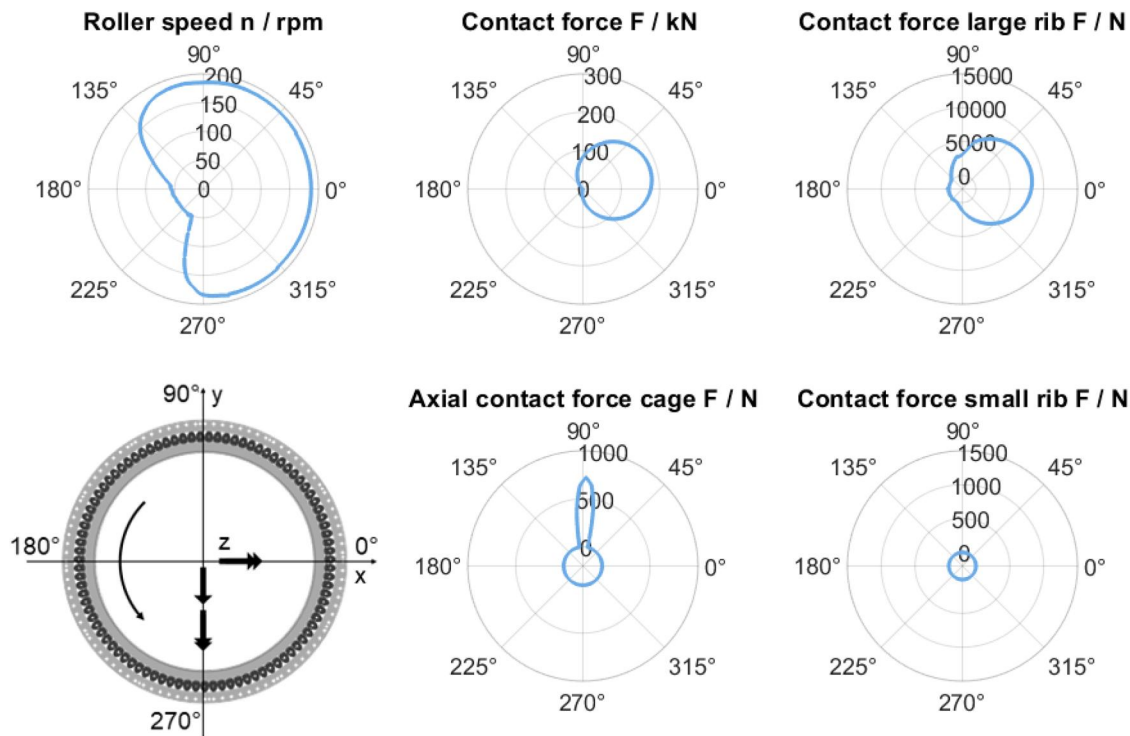


Figure 8. Roller speed, load situation, load zone, contact force between roller and outer rib, contact force between roller and inner rib, and contact force between roller and cage ring (axial contact) at generator side of the main bearing.

both contacts accelerates the roller slightly. A second deceleration occurs immediately afterward due to the rolling element being pressed further against the inner rib by the cage deforming under its own weight. The high amount of friction on the cage ring slows the rolling element down drastically. The rolling element is not loaded by the contact with the raceway and is therefore not held at its speed. This phenomenon is evident in Fig. 6 position (2), where the second deceleration is clearly visible. At position 3, the rolling element enters the load zone. As a result, it is strongly accelerated by the inner ring until it reaches its kinematic speed. Here the rolling element moves outward again on the raceway and comes into contact with the outer rib. However, it is not in axial contact with the cage. The strong acceleration in this area combined with high contact pressure is responsible for the high value of the damage indicator (see Fig. 7). It can therefore be concluded that the high risk of damage results from the huge acceleration when the rolling element enters the load zone. This acceleration takes place because the rolling element was previously decelerated to almost zero rpm. To complete the analysis of the kinematics, we look at position 4 in the bearing. Here, the cage rests on the rolling element again due to its weight and presses it against the outer rib. This causes the rolling element to slow down slightly. However, this effect plays a very minor role and contributes only insignificantly to understanding the damage mechanism. It is therefore not discussed further.

The highest damage risk is found on the rotor side. However, the observed values of the indicator on the generator side are not negligible in LC01. For this reason, the rolling element kinematics are also analyzed on the generator side. Figure 8 shows the roller speed, load distribution

in the bearing, and the contact forces between rolling elements and the ring ribs, as well as the cage.

On the generator side, the rolling elements are decelerated in the range between 110° and 180° . The deceleration is strongest between 150° and 180° . At 320 mW/mm^2 , the damage indicator here is around half as high as during the acceleration on the rotor side. The deceleration is quite rapid. The speed of the rolling element only drops to 50 rpm and therefore not as much as on the rotor side. This is because the rolling element is already in contact with the outer ring again from the 180° position and the rollers are driven by the raceway force. In addition, the cage on this side of the bearing hangs on the rolling elements in the load zone at the top of the bearing and is therefore not supported by the rolling elements outside the load zone. The effect of the rolling elements being clamped between the cage and the ring rib without load does not occur. Consequently, the critical second deceleration phase, as observed on the rotor side, is not present. The rolling elements are then accelerated again between 240° and 270° . This behavior fits very well with the entry into the load zone. The load zone begins at 240° and extends up to 150° . In contrast to the rotor side, the acceleration is lower. The rolling element slip and the damage indicator are less pronounced on this side of the bearing. However, at 400 mW/mm^2 it is still above the specified limit value of 344 mW/mm^2 for this bearing. (28) The highest value of the damage indicator was found on the rotor side in the area of the entry of the load zone. The value was far above the critical value according to the literature. The second peaks of the indicator at the point of slight deceleration at 315° on rotor side is far lower and is not in the range in which damage is to be expected. The detailed

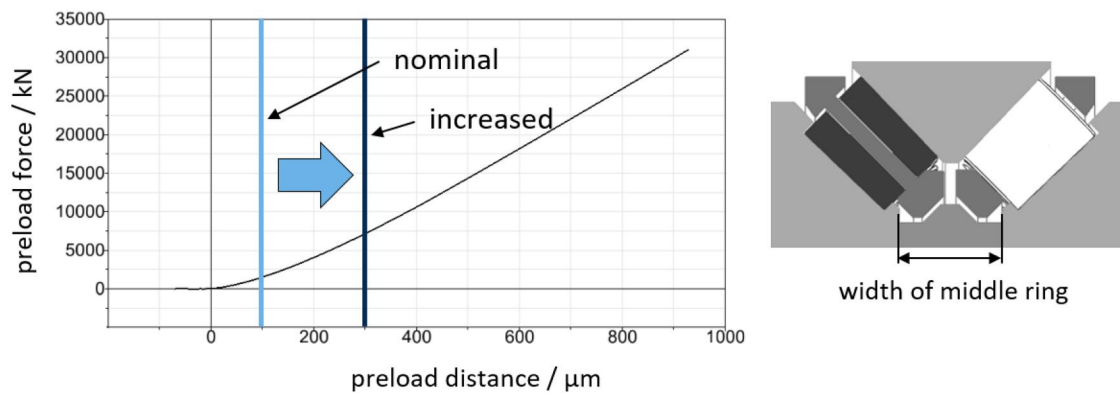


Figure 9. Preload force and preload distance (left) and sectional view of the main bearing with middle ring and preload adjustment option (right).

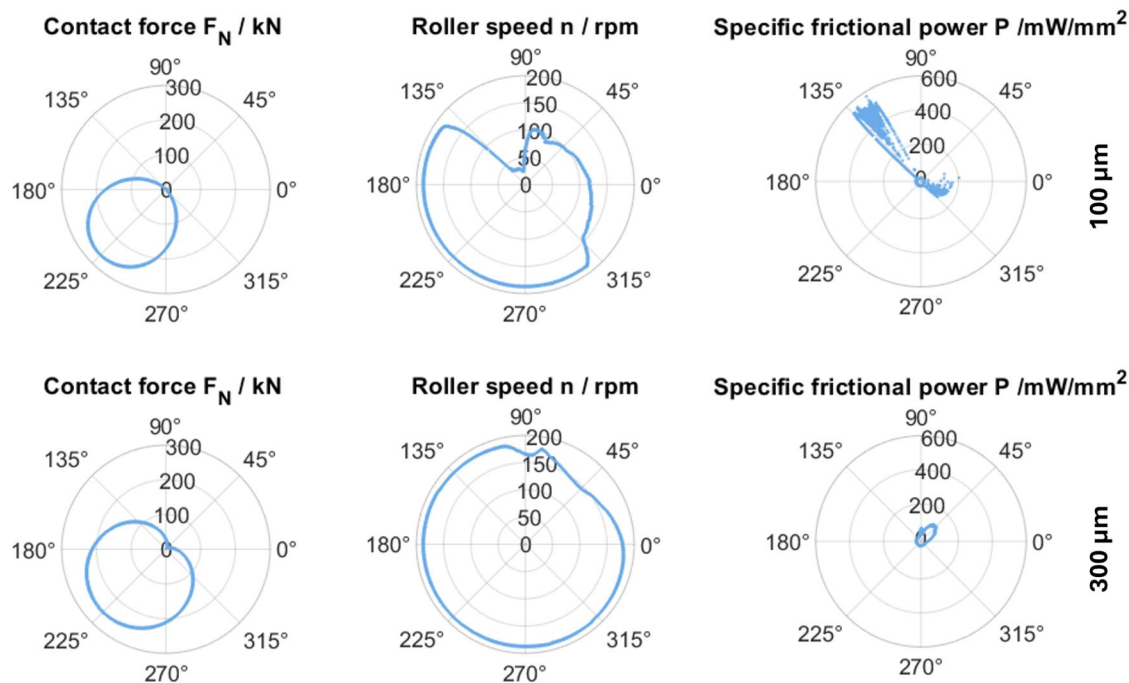


Figure 10. Load zone, roller speed, and specific frictional power at rotor side of the bearing at 100 μm (top) and increased preload of 300 μm (bottom).

analysis of the kinematics and contact forces shows that the highest risk of damage results from a significant deceleration of the rolling element, which becomes trapped outside the load zone between the cage and the small rib of the inner ring, followed by a sharp acceleration when it enters the load zone. In other words, the cause of the harmful slippage is the rolling element being trapped by the deformed cage and the resulting excessive deceleration. It is assumed that the rolling elements will no longer become trapped if they are guided by the raceways through a continuous load zone and no longer become trapped between the cage and the ring rib, especially at position 2 on the rotor side.

Corrective measure

The preload of a bearing is a design parameter. It is calculated in the design process of the drivetrain and is accessible to original equipment manufacturers (OEMs). Previous studies have already shown that the risk of damage due to

smearing can be significantly reduced if the preload is increased. (41,43) Therefore, this study focuses on preload as a solution approach. The right portion of Fig. 9 shows a sectional view of the main bearing under investigation with the middle ring. Its thickness can be used to adjust the preload of the bearing. The left portion of Fig. 9 shows the relationship between preload distance to preload force. The preload of 100 μm and the increased preload of 300 μm are applied.

The increased preload of 300 μm was chosen because values of 200 to 300 μm are common in wind turbines of this size. This solution, however, cannot be applied to the turbine; this can only be done by exchanging the bearing, which means taking the rotor down. The associated costs are easily in the region of one million euros. This highlights the importance of conducting thorough damage risk assessments during the design phase. The influence of the increased preload on load distribution, roller speed, and damage indicator is shown in Fig. 10.

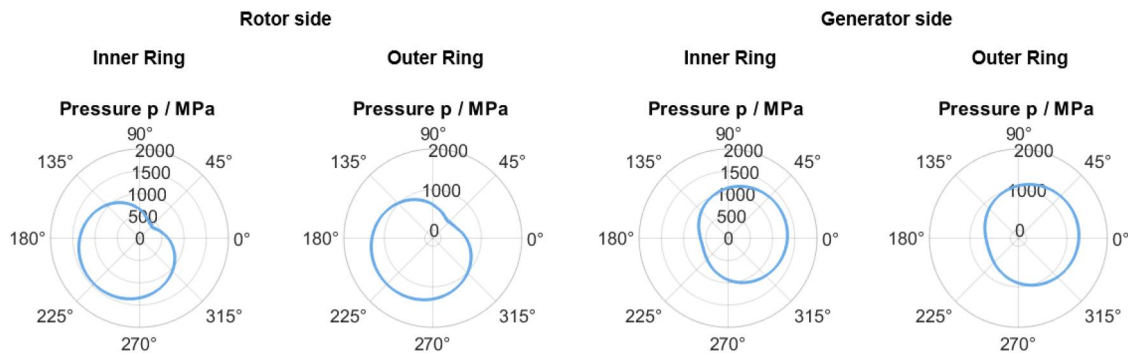


Figure 11. Pressure distribution at increased preload of 300 μm for both roller rows.

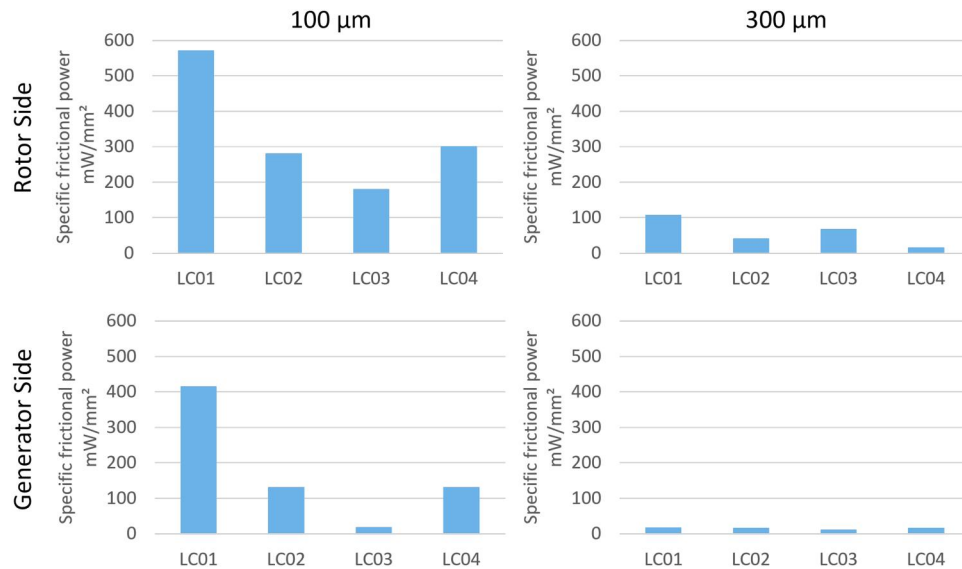


Figure 12. Maxima of smearing risk indicator—specific frictional power—for investigated load cases L01 to L04 at 100 μm (left) and increased preload of 300 μm (right).

It can be clearly seen that the load zone is fully developed after increasing the preload. This means that it now extends over the full circumference of the bearing. The rolling elements are thus loaded during the entire rotation in the bearing. Based on the roller speed shown, it can be seen that the continuous loading of the rolling elements no longer results in the observed strong deceleration in the upper area of the bearing (between 90° and 120°). The rolling elements still decelerate slightly in the range 0° to 40° and then between 80° and 90°, but not nearly as much as before, at 100 μm . The behavior, already explained in the *Analysis* section, remains the same at higher preload of 300 μm . The rolling elements decelerate slightly, are then accelerated briefly, and decelerate a second time. As the load on the rolling elements in this area of the bearing is comparatively low, they can still be accelerated by contact with the ring rib and then decelerated by the cage. Overall, however, the speed of the rolling elements does not fall below 90% of the kinematic speed. Accordingly, the acceleration in the 90° to 100° range is also very moderate. This is no longer the start of the load zone. The rolling elements are nevertheless accelerated, as the load increases sharply in this range. The effect of the low acceleration at higher preload has a very clear effect on the damage indicator. While at 100 μm up to 570 mW/mm^2

is reached during the entry of the rolling elements into the load zone, the value of the indicator at 300 μm is even below 100 mW/mm^2 . The results in Fig. 10 show a significantly reduced damage indicator. Even if the risk of damage does not scale linearly with the indicator according to Hansberg, (28) it is clear that the risk has been significantly reduced.

The increased preload creates an additional load. However, the pressure is still below the fatigue limit stress of 1500 MPa (44) (see Fig. 11) and seems not to be critical.

The same applies to the other three load cases. While the maximum pressure increases slightly in LC03 and LC04, from 1300 MPa to 1320 MPa and from 810 MPa to 1140 MPa, respectively, it actually decreases in LC02, from 1450 MPa to 1350 MPa. Nevertheless, a fatigue life calculation needs to be done in addition.

Influence of the preload increase on standard load cases

In order to evaluate the influence of the increased preload on the damage indicators for smearing risk, the other three load cases were also simulated and analyzed again. The results are shown in Fig. 12.

With the originally set preload of 100 μm (Fig. 12, left), the characteristic values were high on both the rotor side and the generator side. In contrast, for the increased preload of 300 μm (right), these values are comparably low. This is mainly due to the effects described in the *Analysis* section. Enlarging the load zone leads to less deceleration of the rolling elements outside the load zone. As a result, the delta to the kinematic speed of the rolling elements is lower, which means that the acceleration of the rolling elements when entering the load zone, which is essential for the damage characteristic value, is also significantly lower. This means that the risk of damage can be significantly reduced not only in the load case discussed in detail (see *Analysis* section), but also in the other three load cases. A detailed analysis of the kinematics is therefore omitted for the remaining three load cases.

Summary and outlook

In line with previous investigations that proved the applicability of the model, (35–40) the LaMBDA model was used for simulating a main bearing of a wind turbine with a pin-type cage for the first time. The consideration of cage deformation is essential in order to be able to realistically model the mechanisms at work, especially outside the load zone because the deformation of the cage causes the rolling element to be trapped against the rib. This leads to significant deceleration and subsequently harmful slippage when entering the load zone. State-of-the-art approaches for assessing the risk of damage due to smearing can be used as indicators, although the limit values derived in Hansberg (28) are not directly transferable to all bearing sizes. In this study, only short segments of representative load spectra were considered, during which both loads and rotational speed were assumed to remain constant over a short period. For a more comprehensive evaluation of smearing risk, it would be beneficial to simulate transient events such as start-stop cycles or grid failures. Simulating entire load collectives would be also advantageous.

The presented methodology can also be applied by OEMs to perform targeted damage analyses based on field data. Measurements of wind speed and direction provide the basis for the analysis process. In combination with simulation models of the entire system, the loads acting on the main bearing can be determined and thus provide the input data for the dynamic simulation of the bearing. Based on an extensive analysis of the bearing kinematics, it was possible to develop a corrective measure to minimize the risk of damage. In addition, the presented risk assessment procedure can also be integrated into the design process according to IEC standard series 61400 4 (45) and IEC61400-4. (1) This would make it possible to design not only against classical fatigue but also to cover damage mechanisms like adhesive wear. Further characteristic values such as surface-initiated damage according to Koch and Vierendeusel (26) would be included in the evaluation of the simulation results.

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